

# **CSS Filter Properties**

Complete Guide: From Basics to Advanced

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# Chapter 1

## Introduction

### 1.1 What are Filters?

The CSS `filter` property applies graphical effects to elements. You can use filters to blur images, change colors, adjust brightness, add shadows, and much more. Filters work on images, videos, text, and any HTML element.

Think of filters like Instagram or Photoshop effects that you can apply with CSS!

### 1.2 Why Use Filters?

- **No image editing needed:** Modify images on-the-fly with CSS
- **Interactive effects:** Change filters on hover or with JavaScript
- **Performance:** Faster than pre-editing images
- **Responsive:** Apply different filters at different screen sizes
- **Accessibility:** Can be toggled for user preferences

### 1.3 Filter vs. Other Techniques

- **Filter:** Applied to entire element (fast, GPU-accelerated)
- **opacity:** Only changes transparency, not colors
- **blend-modes:** Blends element with background, different purpose
- **box-shadow:** Only adds shadows, limited effects

# Chapter 2

## Basic Filters

### 2.1 blur()

Makes an element blurry or out of focus.

#### 2.1.1 Syntax

```
filter: blur(radius);
```

#### 2.1.2 Parameters

- **radius:** Blur amount in pixels (px). Higher values = more blur
- Default: 0 (no blur)

#### 2.1.3 Examples

##### Slight Blur

```
.image {  
  width: 300px;  
  height: 300px;  
  background-image: url('photo.jpg');  
  background-size: cover;  
  filter: blur(5px);  
}
```

##### Strong Blur Effect

```
.blurry-background {  
  background-image: url('background.jpg');  
  background-size: cover;  
  filter: blur(15px);  
}
```

## Blur on Hover

```
.image {  
  filter: blur(0);  
  transition: filter 0.3s ease;  
}  
  
.image:hover {  
  filter: blur(10px);  
}
```

## 2.2 brightness()

Increases or decreases the brightness of an element.

### 2.2.1 Syntax

```
filter: brightness(percentage);
```

### 2.2.2 Parameters

- **percentage**: 0% = completely dark (black), 100% = original, 150% = brighter
- Default: 100% (original brightness)

### 2.2.3 Examples

#### Make Image Darker

```
.dark-image {  
  width: 300px;  
  height: 300px;  
  background-image: url('photo.jpg');  
  background-size: cover;  
  filter: brightness(60%);  
}
```

#### Make Image Brighter

```
.bright-image {  
  background-image: url('photo.jpg');  
  background-size: cover;  
  filter: brightness(150%);  
}
```

## Brightness on Hover

```
.thumbnail {  
  filter: brightness(80%);  
  transition: filter 0.3s;  
}  
  
.thumbnail:hover {  
  filter: brightness(100%);  
}
```

## 2.3 contrast()

Adjusts the difference between light and dark areas.

### 2.3.1 Syntax

```
filter: contrast(percentage);
```

### 2.3.2 Parameters

- **percentage:** 0% = gray, 100% = original, 150% = high contrast
- Default: 100% (original contrast)

### 2.3.3 Examples

#### Low Contrast (Faded Look)

```
.faded {  
  background-image: url('photo.jpg');  
  background-size: cover;  
  filter: contrast(50%);  
}
```

#### High Contrast (Bold Look)

```
.bold-image {  
  background-image: url('photo.jpg');  
  background-size: cover;  
  filter: contrast(150%);  
}
```

## 2.4 opacity()

Changes the transparency of an element (similar to opacity property but applied as filter).

### 2.4.1 Syntax

```
filter: opacity(percentage);
```

### 2.4.2 Parameters

- **percentage**: 0% = fully transparent, 100% = fully opaque

### 2.4.3 Examples

#### Semi-transparent

```
.transparent-box {  
  width: 200px;  
  height: 200px;  
  background: blue;  
  filter: opacity(50%);  
}
```



# Chapter 3

## Color Filters

### 3.1 grayscale()

Converts colors to grayscale (black and white).

#### 3.1.1 Syntax

```
filter: grayscale(percentage);
```

#### 3.1.2 Parameters

- **percentage:** 0% = original colors, 100% = fully grayscale

#### 3.1.3 Examples

##### Full Grayscale

```
.grayscale-image {  
  width: 300px;  
  height: 300px;  
  background-image: url('photo.jpg');  
  background-size: cover;  
  filter: grayscale(100%);  
}
```

##### Partial Grayscale

```
.muted-colors {  
  background-image: url('photo.jpg');  
  background-size: cover;  
  filter: grayscale(50%);  
}
```

### Grayscale on Hover (Reveal Colors)

```
.gallery-item {  
  filter: grayscale(100%);  
  transition: filter 0.3s;  
}  
  
.gallery-item:hover {  
  filter: grayscale(0%);  
}
```

## 3.2 sepia()

Gives a warm, brownish (vintage) tone to images.

### 3.2.1 Syntax

```
filter: sepia(percentage);
```

### 3.2.2 Parameters

- **percentage:** 0% = original, 100% = full sepia tone

### 3.2.3 Examples

#### Full Sepia (Vintage Look)

```
.vintage-photo {  
  width: 300px;  
  height: 300px;  
  background-image: url('old-photo.jpg');  
  background-size: cover;  
  filter: sepia(100%);  
}
```

#### Subtle Sepia

```
.warm-image {  
  background-image: url('photo.jpg');  
  background-size: cover;  
  filter: sepia(30%);  
}
```

## 3.3 saturate()

Increases or decreases color saturation (vividness of colors).

### 3.3.1 Syntax

```
filter: saturate(percentage);
```

### 3.3.2 Parameters

- **percentage**: 0% = completely desaturated (grayscale), 100% = original, 200% = super vibrant

### 3.3.3 Examples

#### Desaturated Colors

```
.muted-image {  
  background-image: url('photo.jpg');  
  background-size: cover;  
  filter: saturate(0%);  
}
```

#### Vibrant Colors

```
.vibrant-image {  
  background-image: url('photo.jpg');  
  background-size: cover;  
  filter: saturate(200%);  
}
```

---

## 3.4 hue-rotate()

Rotates colors around the color wheel.

### 3.4.1 Syntax

```
filter: hue-rotate(degrees);
```

### 3.4.2 Parameters

- **degrees**: Rotation angle (0deg = original, 360deg = back to original)

### 3.4.3 Examples

#### Shift Colors 90 Degrees

```
.shifted-colors {  
  width: 300px;  
  height: 300px;  
  background-image: url('photo.jpg');  
  background-size: cover;  
  filter: hue-rotate(90deg);  
}
```

### Invert Colors (180 Degrees)

```
.inverted {  
  background-image: url('photo.jpg');  
  background-size: cover;  
  filter: hue-rotate(180deg);  
}
```

### Animated Color Rotation

```
.rainbow-effect {  
  background-image: url('photo.jpg');  
  background-size: cover;  
  animation: rotate-hue 3s infinite;  
}  
  
@keyframes rotate-hue {  
  0% { filter: hue-rotate(0deg); }  
  100% { filter: hue-rotate(360deg); }  
}
```

---

## 3.5 invert()

Inverts all colors (like a photographic negative).

### 3.5.1 Syntax

```
filter: invert(percentage);
```

### 3.5.2 Parameters

- **percentage**: 0% = original, 100% = fully inverted

### 3.5.3 Examples

#### Full Color Inversion

```
.inverted-image {  
  width: 300px;  
  height: 300px;  
  background-image: url('photo.jpg');  
  background-size: cover;  
  filter: invert(100%);  
}
```

### Partial Inversion

```
.weird-effect {  
  background-image: url('photo.jpg');  
  background-size: cover;  
  filter: invert(50%);  
}
```

## Chapter 4

# Shadow & Effect Filters

### 4.1 drop-shadow()

Adds a shadow effect to the element itself (not the box).

#### 4.1.1 Syntax

```
filter: drop-shadow(offset-x offset-y blur-radius color);
```

#### 4.1.2 Parameters

- **offset-x**: Horizontal shadow offset
- **offset-y**: Vertical shadow offset
- **blur-radius**: How soft the shadow is
- **color**: Shadow color (optional, defaults to black)

#### 4.1.3 Examples

##### Simple Shadow

```
.element {  
  width: 200px;  
  height: 200px;  
  background: blue;  
  filter: drop-shadow(5px 5px 10px black);  
}
```

##### Colored Shadow

```
.box {  
  width: 200px;  
  height: 200px;  
  background: red;  
  filter: drop-shadow(0px 10px 20px rgba(0, 0, 0, 0.5));  
}
```

```
}
```

### Shadow on Text

```
.text {
  font-size: 48px;
  color: white;
  filter: drop-shadow(3px 3px 6px rgba(0, 0, 0, 0.8));
}
```

### Multiple Shadows (Layered)

```
.element {
  filter: drop-shadow(2px 2px 4px red)
        drop-shadow(4px 4px 8px blue);
}
```

## 4.2 Combining Multiple Filters

You can combine multiple filters in one property!

### 4.2.1 Syntax

```
filter: blur(5px) brightness(80%) contrast(150%);
```

### 4.2.2 Examples

#### Professional Photo Effect

```
.professional {
  width: 300px;
  height: 300px;
  background-image: url('photo.jpg');
  background-size: cover;
  filter: brightness(105%)
        contrast(110%)
        saturate(120%);
}
```

#### Dreamy Effect

```
.dreamy {
  background-image: url('photo.jpg');
  background-size: cover;
  filter: blur(3px)
```

```
brightness(110%)
saturate(80%);
}
```

### Dark Moody

```
.dark-moody {
  background-image: url('photo.jpg');
  background-size: cover;
  filter: brightness(70%)
         contrast(130%)
         saturate(110%);
}
```

### Vintage Film

```
.vintage {
  background-image: url('photo.jpg');
  background-size: cover;
  filter: sepia(40%)
         brightness(95%)
         contrast(120%);
}
```



## Chapter 5

# Real-World Use Cases

### 5.1 Image Gallery with Hover Effects

```
<style>
  .gallery {
    display: grid;
    grid-template-columns: repeat(auto-fit,
                                   minmax(200px, 1fr));
    gap: 20px;
  }

  .gallery-item {
    width: 100%;
    aspect-ratio: 1 / 1;
    background-size: cover;
    filter: grayscale(80%) brightness(80%);
    transition: filter 0.3s ease;
    cursor: pointer;
  }

  .gallery-item:hover {
    filter: grayscale(0%) brightness(100%);
  }
</style>

<div class="gallery">
  <div class="gallery-item"
    style="background-image: url('img1.jpg')"></div>
  <div class="gallery-item"
    style="background-image: url('img2.jpg')"></div>
</div>
```

---

### 5.2 Team Member Cards

```
<style>
  .team-card {
```

```
    width: 200px;
    text-align: center;
}

.team-card img {
    width: 100%;
    height: 200px;
    border-radius: 10px;
    filter: brightness(100%) contrast(110%);
    transition: filter 0.3s;
}

.team-card:hover img {
    filter: brightness(110%) contrast(120%);
}

.team-card h3 {
    margin-top: 10px;
    font-size: 20px;
}
</style>

<div class="team-card">
  
  <h3>John Smith</h3>
</div>
```

## 5.3 Product Showcase

```
<style>
.product {
    position: relative;
    width: 300px;
    overflow: hidden;
}

.product img {
    width: 100%;
    filter: brightness(95%) saturate(110%);
    transition: filter 0.4s ease;
}

.product:hover img {
    filter: brightness(110%) saturate(130%);
    transform: scale(1.05);
}

.product-info {
    padding: 15px;
    background: white;
}
```

```

</style>

<div class="product">
  
  <div class="product-info">
    <h4>Premium Headphones</h4>
    <p>Price: \$99.99</p>
  </div>
</div>

```

## 5.4 Dark Mode Background

```

<style>
  body.dark-mode {
    background: #1a1a1a;
    color: #fff;
  }

  body.dark-mode .background-image {
    filter: brightness(40%) contrast(120%);
    background-attachment: fixed;
  }

  body.light-mode .background-image {
    filter: brightness(100%) contrast(100%);
  }
</style>

<button onclick="toggleDarkMode()">Toggle Dark Mode</button>

<script>
  function toggleDarkMode() {
    document.body.classList.toggle('dark-mode');
  }
</script>

```

## 5.5 Animated Filter Effects

```

<style>
  .animated-box {
    width: 200px;
    height: 200px;
    background: linear-gradient(45deg, red, blue);
    animation: filter-animate 3s infinite;
  }

  @keyframes filter-animate {

```

```
    0% {  
      filter: brightness(100%) hue-rotate(0deg);  
    }  
    50% {  
      filter: brightness(150%) hue-rotate(180deg);  
    }  
    100% {  
      filter: brightness(100%) hue-rotate(360deg);  
    }  
  }  
</style>  
  
<div class="animated-box"></div>
```

## Chapter 6

# Browser Support & Best Practices

### 6.1 Browser Support

CSS filters are widely supported:

- Chrome 18+
- Firefox 35+
- Safari 6+
- Edge 12+
- Mobile browsers (iOS Safari 6+, Chrome Mobile)

Note: For older browsers, provide fallback images or alternative styling.

---

### 6.2 Performance Tips

1. **GPU Acceleration:** Filters are GPU-accelerated, but avoid excessive use on many elements
2. **will-change:** Use sparingly for animated filters

```
.filtered-element {  
  will-change: filter;  
}
```

3. **Optimize Images:** Use optimized images; filters don't improve poor quality
  4. **Mobile Performance:** Test filters on mobile devices; some devices may struggle with complex filters
-

## 6.3 Common Gotchas

### 6.3.1 Filter vs. Opacity

- **filter: opacity()** applies filters to the element
- **opacity property** affects stacking context and children
- Use filters when you want CSS effects; use opacity for transparency

### 6.3.2 Combining Filters Order Matters

```
/* Different order, different visual result */
filter: brightness(50%) saturate(200%);
filter: saturate(200%) brightness(50%);
```

### 6.3.3 Filters Don't Affect ::before/::after

- Filters applied to parent don't affect pseudo-elements positioned outside
- Apply filters to pseudo-elements individually if needed

—

## 6.4 Accessibility Considerations

- **Prefers-reduced-motion:** Avoid animated filters for users with motion sensitivity

```
@media (prefers-reduced-motion: reduce) {
  .animated { animation: none; }
}
```

- **Color Contrast:** Ensure filters don't reduce text readability
- **Brightness for Important Content:** Don't make critical content too dark with brightness filters

# Chapter 7

## Quick Reference Guide

### 7.1 All Filter Functions

- **blur(px)**: Blurs the element (5px, 10px, etc.)
  - **brightness(%)**: Brightens/darkens (50%, 100%, 150%)
  - **contrast(%)**: Adjusts contrast (50%, 100%, 150%)
  - **grayscale(%)**: Converts to grayscale (0% = color, 100% = gray)
  - **sepia(%)**: Vintage warm tone (0% = normal, 100% = sepia)
  - **saturate(%)**: Color intensity (0% = gray, 100% = normal, 200% = vivid)
  - **hue-rotate(deg)**: Rotates color wheel (0deg to 360deg)
  - **invert(%)**: Inverts colors (0% = normal, 100% = inverted)
  - **opacity(%)**: Transparency (0% = invisible, 100% = opaque)
  - **drop-shadow()**: Adds shadow effect
- 

### 7.2 Common Combinations

#### 7.2.1 Instagram-Style Filters

```
/* Lo-Fi */
filter: brightness(90%) contrast(85%) saturate(110%);

/* Vintage */
filter: sepia(30%) brightness(100%) saturate(110%);

/* High Contrast */
filter: contrast(130%) saturation(120%);

/* Dreamy */
filter: blur(2px) brightness(110%) saturate(90%);
```

---

## 7.3 Filters for Different Elements

### 7.3.1 Images

```
img { filter: brightness(110%) contrast(120%); }
```

### 7.3.2 Videos

```
video { filter: saturate(120%) hue-rotate(10deg); }
```

### 7.3.3 Text

```
h1 { filter: drop-shadow(2px 2px 4px rgba(0,0,0,0.5)); }
```

### 7.3.4 Backgrounds

```
.section {  
  background-image: url('bg.jpg');  
  filter: brightness(60%) blur(3px);  
}
```